TECHNION | The Henry and Marilyn Taub Faculty of Computer Science

Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 2a - Actuation and Agency

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Accounting for Actions and Partial Observability

So far, we have considered systems that evolve over time.

These models describe **passive stochastic processes**.

Many real systems include an **agent** that can influence the system.

In this module, we extend the model in two ways:

1. Adding actuation
2. Accounting for partial observability

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Various Notations

This course fuses ideas from different research communities: learning, RL, planning, control, etc.

Each research community has its own set of notations. We will use various notations interchangeably.

Examples:

- state at time t can be represented as x_t or s_t
- observation at t is o_t or z_t.
- sometimes, evidence e_t is used to represent the sequence of observations, or observation-action pairs, or observation-action-rewards tuples.
- evidence $e_{1:t}$ collected over time is often called a *history* h.
- action at t is a_t or $\mathbf{u}_t$.

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Actions, Rewards, Policies

Passive Stochastic Dynamics

A stochastic dynamical system evolves according to some (possibly unknown) model

$$X_{t+1} \sim \Pr(\cdot \mid X_t)$$

Characteristics

- The system evolves over time
- Transitions may be stochastic
- No agent controls the process

Examples:

- Weather evolution
- Population dynamics
- Electricity prices

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Adding Actuation

Now we have actions that can be applied at each time step

$$A_t \in \mathcal{A}$$

The system dynamics now also depend on the action:

$$\Pr(X_{t+1} \mid X_t, A_t)$$

The system is a **controlled stochastic process** or **controlled stochastic dynamical system**.

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Fixed Policy Assumption

This course is about coming up with *good* policies, but for now we assume actions are generated by a fixed policy π.

With a fixed policy, the challenge is to understand how the system evolves when the policy is followed.

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Controlled Stochastic Dynamics

The system evolves according to

$$X_{t+1} \sim \Pr(\cdot \mid X_t, A_t)$$

with actions generated by the policy

$$A_t \sim \pi(\cdot)$$

The policy determines which actions occur, but the transition model determines their effect.

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Actions

Actions can be defined in various ways:

- **Deterministic / non-deterministic / stochastic**
- **Continuous / discrete**
- From **high-level commands** (pick-up apple, make breakfast) to **low-level control signals** (joint torques, motor commands)
- May have **different durations or execution uncertainty**

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Policy

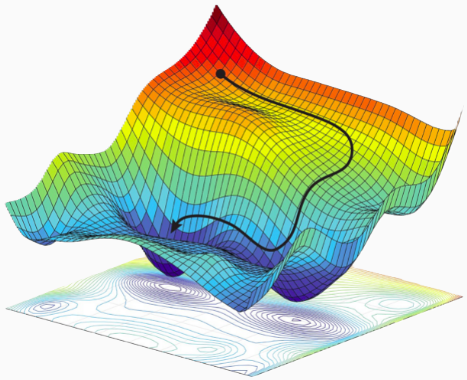
- A **deterministic** policy is a mapping:
  - $\pi : \mathcal{S} \rightarrow \mathcal{A}$ from states to actions
  - $\pi : \mathcal{B} \rightarrow \mathcal{A}$ from beliefs to action
  - $\pi(s)$/ $\pi(\beta)$ is the (single) action the agent will perform at state s or belief β.
- A **non-deterministic** policy is a mapping:
  - $\pi : \mathcal{S} \rightarrow \mathcal{A}^n$ from states to actions,
  - $\pi : \mathcal{B} \rightarrow \mathcal{A}^n$ from beliefs to actions
  - $\pi(s)$/ $\pi(\beta)$ is the set of actions, one of which the agent will perform at state s or belief β.
- A **stochastic** policy is a mapping:
  - $\pi : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ from states to actions
  - $\pi : \mathcal{B} \times \mathcal{A} \rightarrow [0, 1]$ from beliefs to action
  - $\pi(s, a)$/ $\pi(\beta, a)$ is the probability the agent will perform action a at state s/ belief β.

Why do we need a probabilistic policy?

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Policy Parameters

- Instead of using a policy $\pi(a|s)$ that explicitly maps states (or beliefs to actions), we can use a **parameterized policy** $\pi_\theta(a|s)$, i.e., as a function with parameters θ.
- For deterministic policies θ specify the action taken at each state, and for stochastic policies, they induce a probability distribution over actions.
- Examples:
  - Linear policies $\pi_\theta(s) = \theta^\intercal s$ where θ is a vector of weights.
  - Neural network policies where θ are the weights of the network.



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Actions

Actions $\neq$ Agency

Introducing actions does not imply that an agent is making decisions.

- Actions can simply be **inputs applied to the system**.
- The actions may be generated by a **fixed controller or policy**.
- The challenge is to model how actions **influence the system dynamics** via $\Pr(X_{t+1} \mid X_t, A_t)$.

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Reward

A reward function assigns a numerical value to the outcome of an interaction with the environment.

Rewards are part of the **environment model** and describe the feedback received by the agent.

They play a central role in defining the **objective** that the agent will eventually try to optimize.

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Representing Reward

There are various ways to represent reward.

With a **deterministic reward** $\mathcal{R}(s, a, s') \in \mathbb{R}$, once the transition $S_t = s, A_t = a, S_{t+1} = s'$ is fixed, the reward value is fixed.

In its most general form, the reward is a **stochastic variable**:

$$R_{t+1} \sim \mathcal{R}(\cdot \mid S_t = s, A_t = a, S_{t+1} = s')$$

Equivalently, the **reward distribution** can be written as

$$\Pr(R_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s')$$

A deterministic representation of the **expected reward**:

$$R^a_{s,s'} = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s']$$

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Representing Reward (contd.)

Representing reward as a function

$$\mathcal{R}(s, a, s')$$

Often, rewards are defined over origin states $\mathcal{R}(s)$, destination states $\mathcal{R}(s')$, or over state-action pairs $\mathcal{R}(s, a)$.

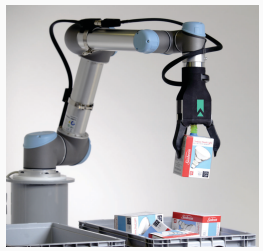
Sometimes, rewards are induced by the environment (e.g., fuel costs), and sometimes they are designed (shaped) to encode how desirable an outcome is. This is called Reward Shaping.

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Reward Functions Examples

Robotics

$$\mathcal{R} = \begin{cases} +1 & \text{successful grasp} \\ 0 & \text{failure} \end{cases}$$



Games

$$\mathcal{R} = \text{score gained at step } t$$



Control / Energy

$$\mathcal{R} = -\text{energy consumption}$$



What is the reward function encoding for each setting?

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Deep Dive

David Silver, Satinder Singh, Doina Precup, Richard S. Sutton

Reward is enough

Artificial Intelligence, Volume 299, 2021

https://www.sciencedirect.com/science/article/pii/S0004370221000862

https://www.youtube.com/watch?v=u5hBHkbihxA

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From the Literature: Reward is enough?

Key claim: Many abilities associated with intelligence may arise from **maximizing reward signals**.

Examples of capabilities that have emerged from reward-driven learning:

- game playing (AlphaGo, AlphaZero)
- robotics control
- language models trained with feedback
- complex planning and strategy

A system that learns to maximize reward in rich environments may develop many components of intelligence.

Are there aspects of intelligence that cannot be captured by reward maximization?

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Return

Reward R: immediate feedback
Return G: total future reward

Key idea: future rewards are often **discounted** using a factor

$$0 \leq \gamma < 1$$

This reflects both:

- a natural preference for rewards received sooner rather than later, and
- a computational advantage, since discounting helps ensure convergence.

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Return

- Expected return for episodic tasks:

$$G_t = R_{t+1} + R_{t+2} + R_{t+3} + ... + R_T = \sum_{k=0}^{T} R_{t+k+1}$$

- Expected return for continuing (infinite) tasks:

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + ... = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

- Returns at successive time steps are related:

$$G_t = R_{t+1} + \gamma(R_{t+2} + \gamma R_{t+3} + \gamma^2 R_{t+4} + \ldots) = R_{t+1} + \gamma G_{t+1}$$

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Value Functions

Return quantifies the **total future reward**.

Value functions estimate how **good** it is:

- to be in a state
- to take an action in a state

Key idea: "goodness" is defined as **expected return**.

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Value Functions

Since the rewards the agent can expect to receive in the future depend on what actions it will take, value functions are defined with respect to particular policies.

State-Value Function for Policy π:

$$V_\pi(s) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s \right]$$

State-Action-Value Function for Policy π:

$$Q_\pi(s, a) \doteq \mathbb{E}_\pi \left[G_t \mid S_t = s, A_t = a \right]$$

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True vs. Estimated Values - Notations

True (Unknown) value functions are denoted as:

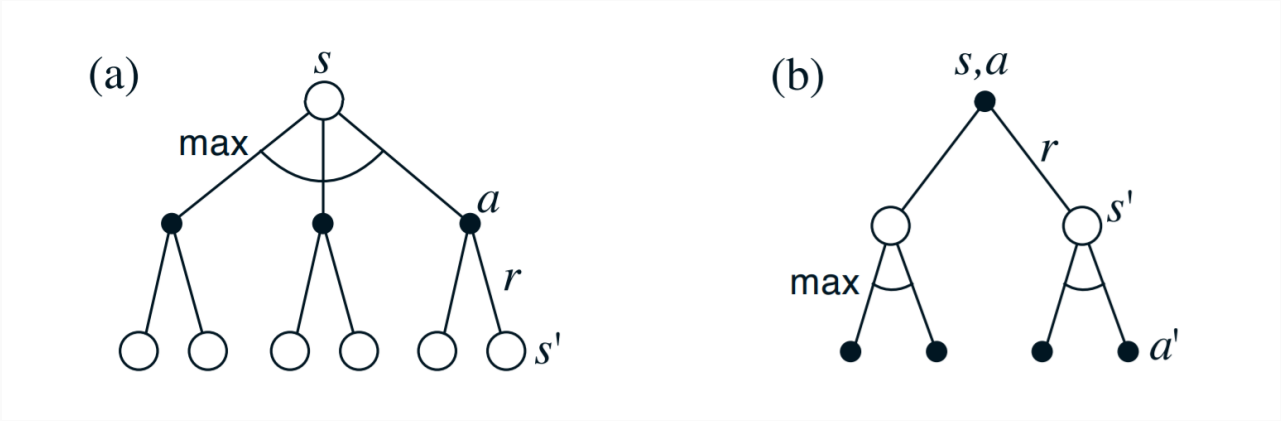
- $V_\pi(s)$ — true state value
  - Defined via expectation: $V_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$
- $Q_\pi(s, a)$ — true action value
  - Defined via expectation: $Q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$

Estimated (learned) value functions (updated over time) are denoted as:

- $\hat{V}_\pi(s)$ — estimated state value
- $\hat{Q}_\pi(s, a)$ — estimated state-action value

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Value Functions



Backup diagrams for $V_\pi(s)$ (left) and $Q_\pi(s,a)$ (right).

Since it may not be practical to keep separate averages for each state individually V_π and Q_π are often represented as parameterized functions (with fewer parameters than states).

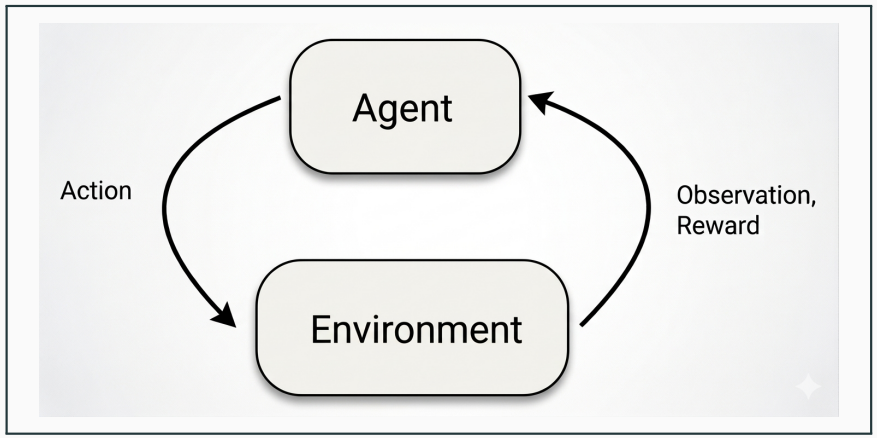
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The Agent-Environment Interface

The Agent-Environment Interface

An agent operates in the environment by taking actions.

An agent (and its behavior) is characterized by its **objectives** and **capabilities**.



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The Agent-Environment Interface

Typically, a model for AI decision-making comprises the following components:

- state space
- sensor function
- action space
- reward function
- horizon
- objective

A model is an **abstraction** of the underlying environment.

Each model can be used by different decision-making procedures.

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The Agent-Environment Interface

The agent interacts with the environment at discrete time steps $t = 0, 1, 2, \ldots$.

At each time step t, the agent observes the current state $S_t \in \mathcal{S}$ [1] and selects an action $A_t \in \mathcal{A}(S_t = s)$.

As a result of applying A_t at S_t, at the next time step $t + 1$, the agent receives a numerical reward R_{t+1} and transitions to a new state S_{t+1} based on the dyanmics model

$$S_{t+1} \sim \Pr(\cdot \mid S_t, A_t)$$

The agent and environment together thereby give rise to a sequence or **trajectory**: $\tau = S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \ldots$

[1] recall that we can use states as $\mathcal{X}$ instead

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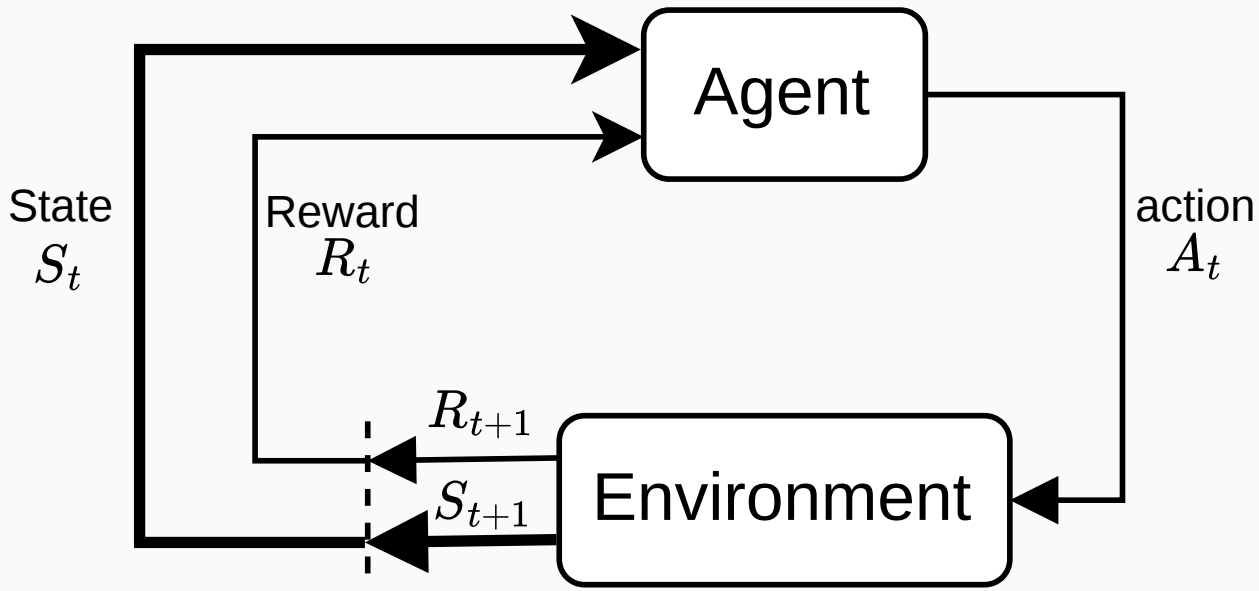
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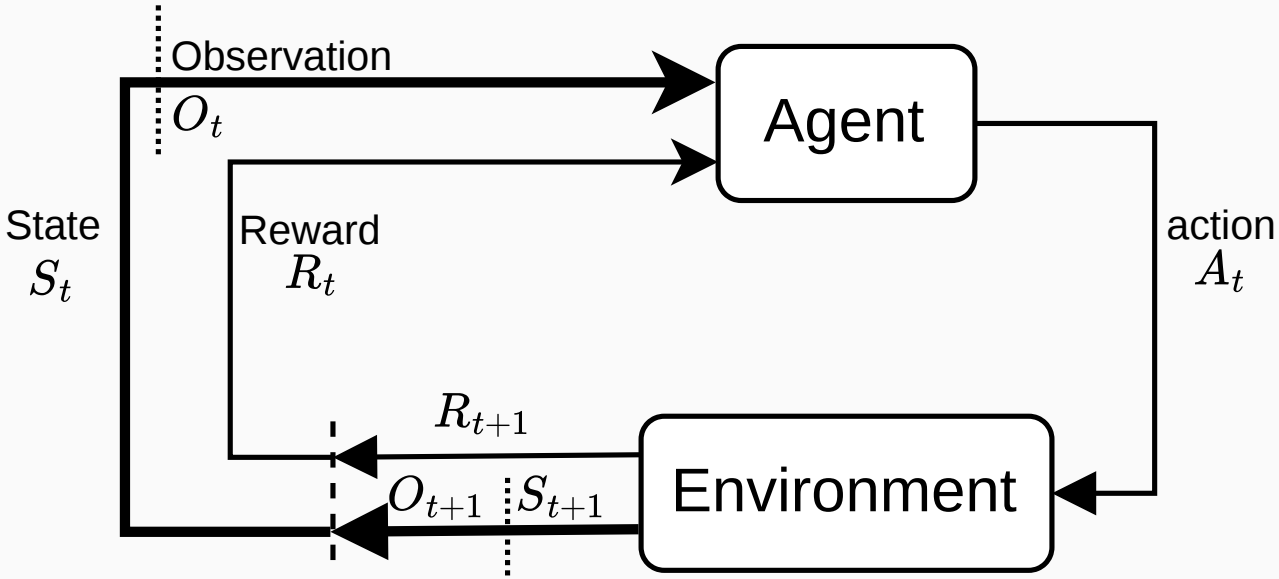
Partial Observability

The Agent-Environment Interface



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The Agent-Environment Interface Under Partial Observability



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The Agent-Environment Interface

- **Episodic tasks:** When the agent–environment interaction breaks naturally into sub-sequences (e.g., limited time or resources). Each episode ends in a special state called the *terminal state*.
- **Continuing tasks:** When the agent–environment interaction does not break naturally into identifiable episodes, but goes on continually without limit.

Real-world examples of episodic vs. continuing tasks ?

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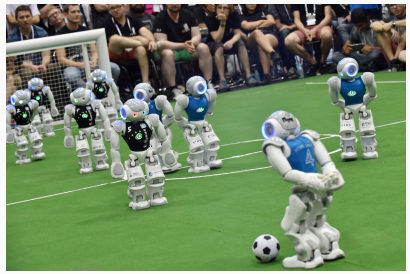
So far, we have considered the different elements that are used to model the environment, capturing its dynamics, uncertainty, etc.

We also considered **actuation** and **actions** that affect the environment, and **policies** that prescribe which actions are applied at a given state or belief.

We now consider **agency** - the ability to decide how to act, i.e., the process of computing policies.

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Reminder: Reactive vs. Rational Agent

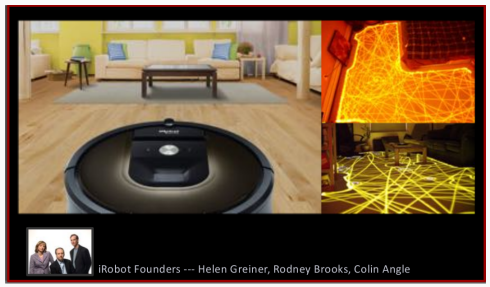



- **Reactive** or **Reflex** agent:
  - simple reactions to stimulus (short-term)
  - fast response times and limited resource usage
- **Rational agent:**
  - extended reasoning capabilities
  - can achieve complex objectives
  - utility maximizing
  - can learn new knowledge and react to change

We focus on rational agents.

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Reactive Agents



SIMPLE ROOMBA

If BUMP = TRUE
THEN Turn (random direction/amt)

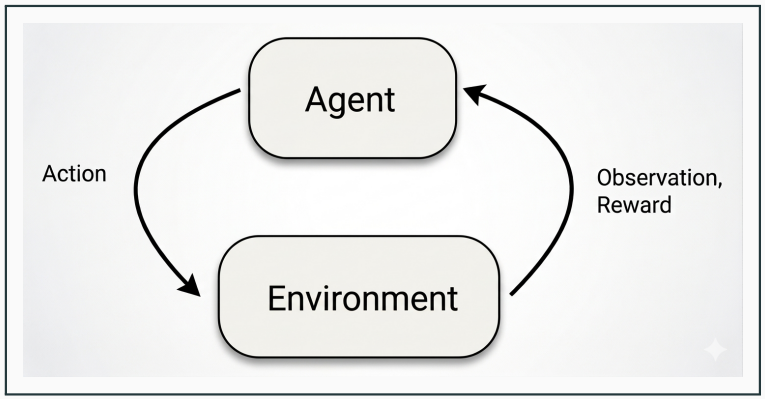
ELSE MOVE-STRAIGHT

https://www.youtube.com/watch?v=7FSUtSurqA4

We focus on rational agents.

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An Agent



An agent does not have a pre-defined policy.
It has agency to decide how to act.

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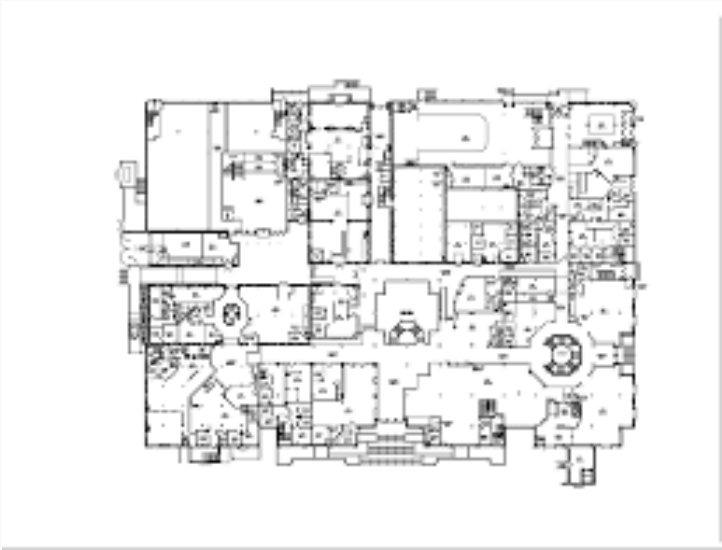
Policies for Rational Agents

- Typically cannot be described explicitly for each state and require a compact representation.
- An **optimal policy**, denoted π^* is *better* than or equal to all other policies.
- Since our focus is on rational agents with a clearly defined utility measure, an optimal policy maximizes the agent's utility function $\mathcal{U}$.



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How to come up with ’good’ policies ?





Sarah Keren

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Why Sequential Decision-Making?

Many real-world problems require reasoning **beyond immediate outcomes**.

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Why Sequential Decision-Making?

Many real-world problems require reasoning **beyond immediate outcomes**.

- **Information gathering affects future decisions**
  - An agent may choose to sense the environment before acting.
- **Actions influence future opportunities**
  - An agent may take a longer path now to avoid obstacles later.
- **Short-term rewards may conflict with long-term goals**
  - Energy systems may incur temporary costs to maintain grid stability.
- **Decisions accumulate over time**
  - Poor early choices can lead to irreversible states.

Sequential decision-making provides a framework for optimizing behavior over time rather than step-by-step reactions.

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Many Approaches to Control

Sequential decision-making in AI is not the only framework used to control dynamical systems. Many approaches have been developed in other research disciplines.

In robotics, for example, classical **feedback controllers** (e.g., PID) are designed to react to the current system state without explicitly planning far into the future.
In contrast, **optimal control** includes methods for computing a sequence of control inputs $u_0, u_1, \ldots, u_T$ that minimizes a cost function over time while satisfying the system dynamics.

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Many Approaches to Control*

One widely used approach is **Model Predictive Control (MPC)**, which repeatedly solves a short-horizon optimization problem and applies only the first control action before replanning.

Example: An autonomous car approaching a curve computes a steering and acceleration plan for the next few seconds. It executes only the first control input, observes the new state of the vehicle, and then recomputes the plan.



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Sequential Decision-Making

Our focus in this course is on **sequential decision-making**, as studied under **AI and reinforcement learning (RL)**.

In these settings, an agent must choose actions **over time**, where each action affects the future state of the environment.

We will mostly assume the state space is **discrete and finite**, although it may be extremely large.

This contrasts with approaches where the system dynamics are often modeled in **continuous spaces** and a major challenge is computing or approximating good **discretizations**.

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A note on notations

When we modeled the system as a **stochastic process**:

- State is a random variable: $X_t \in \mathcal{X}$
- Focus: probabilistic evolution and inference

$$\Pr(X_{t+1} \mid X_t), \qquad \Pr(O_t \mid X_t)$$

When considering **decision-making**:

- A state is typically represented as: $s_t \in \mathcal{S}$
- The transition probabilities are represented by a **transition function**:

$$\mathcal{P}(s_t, a, s_{t+1}) = \Pr(X_{t+1} = s' \mid X_t = s, A_t = a)$$

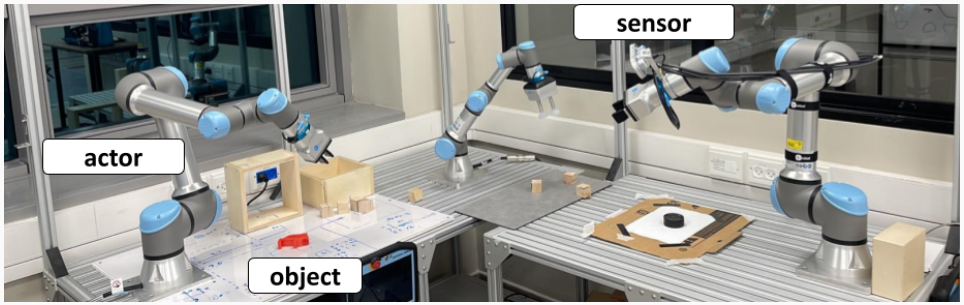
- Observation probabilities are represented by a **sensor function**:

$$\mathcal{O}(o \mid s) = \Pr(O_t = o \mid X_t = s)$$

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Formal Models for Sequential Decision-Making







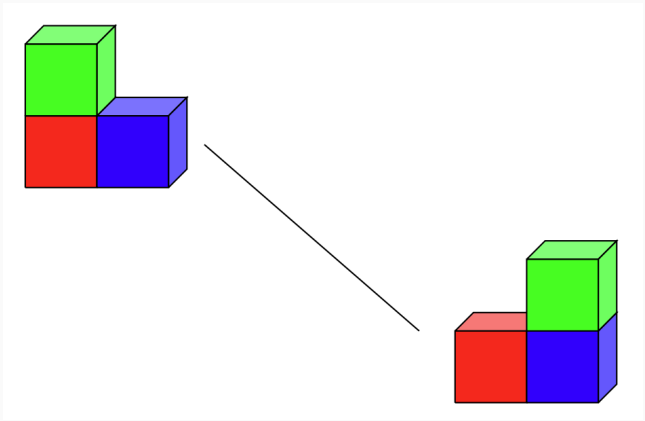
How to model these settings? What are the states, actions, rewards, and observations? What is the objective?

- Route selection (from Arad to Bucharest)
- Solving 15-puzzle (or Rubik's cube, or ...)
- Selecting and ordering movements of an elevator or a crane
- Production lines control
- Autonomous robots
- Crisis management
- ...

What do all these have in common?

Planning and sequential decision making

- All these problems deal with **action selection** or **control**
- Some notion of problem state
- (Often) specification of initial state and/or goal state
- Legal moves or actions that allow transition from states to other states



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Modeling the Environment

What do we need to model ?

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Modeling the Environment

What do we need to model ?

- Sequential decision making
- Uncertainty in action outcomes
- Partial observability

Additional considerations:

The model needs to be compact enough to be solvable but informative enough to be useful.

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Representation Complexity

The **representation complexity** of a model is the amount of information required to specify it.

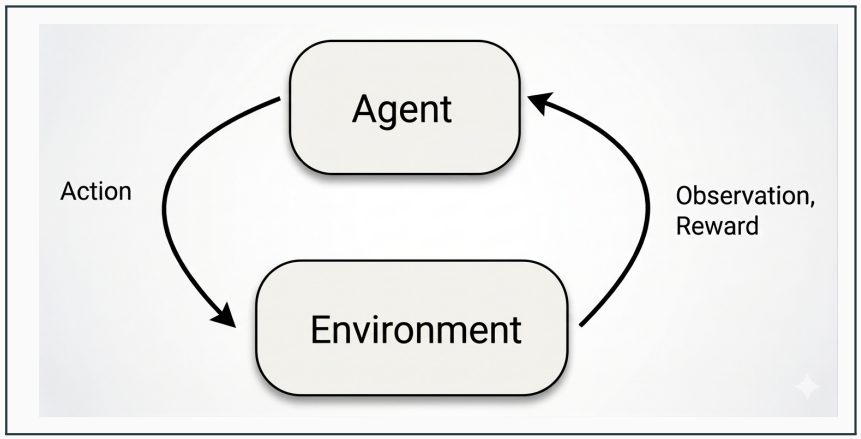
Formally, for a model class $\mathcal{M}$ and a particular model $M \in \mathcal{M}$, the representation complexity of M is the number of parameters, entries, or bits needed to encode its components, i.e., its states, actions, transitions, observations, rewards, ...

It captures the size of the *input description* of the decision-making problem, as opposed to the time needed to solve it (which we will discuss when considering solution approaches).

Key challenge: model rich enough to be useful vs. simple enough to solve

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Modeling the Environment



Is the step-by-step model always a reasonable assumption?

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Different classes of problems

Properties

- **Dynamics:** deterministic, nondeterministic or probabilistic
- **Observability:** full, partial, or none
- **Horizon:** finite or infinite
- ...

Side comment ...

- It is not that deterministic problems are easy
- It is not even clear that they are too far from modeling real-world problems well!

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Properties of the setting: dynamics

Deterministic dynamics

Action + current state uniquely determine successor state.

Nondeterministic dynamics

For each action and current state there may be several possible successor states.

Probabilistic dynamics

For each action and current state there is a probability distribution over possible successor states.

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The Agent and its Decision Making Problem

An agent typically includes one or more of these components:

- **Model:** agent's representation of the environment:
- **Value function:** capture how good is each state and/or action
- **Policy:** agent's behaviour function

We will consider three environment models of increasing complexity:

- Deterministic fully observable environments
- Stochastic fully observable environments
- Stochastic partially observable environments

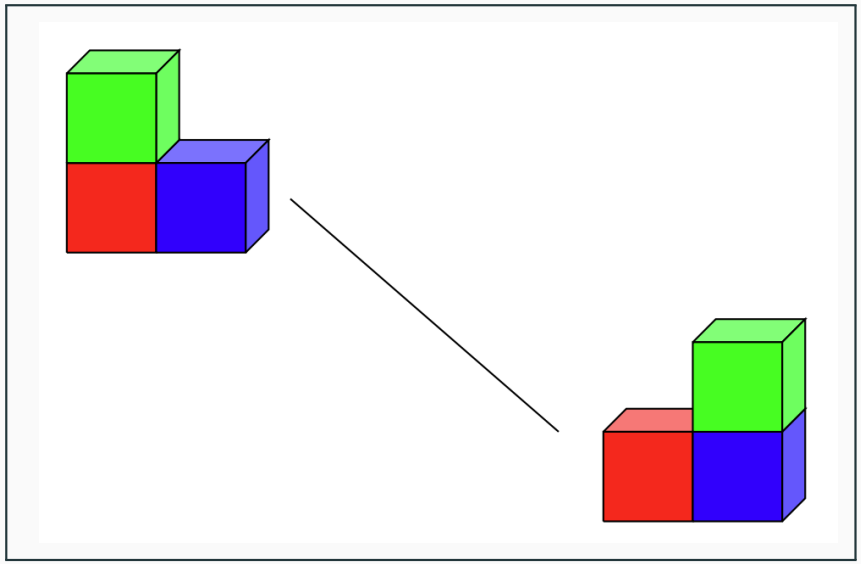
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Deterministic Fully-Observable Domains

Deterministic dynamics example

Pick and Place: Moving objects with a robotic hand:

```
stack(green_block, blue_block)
```



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Propositional STRIPS

Deterministic fully observable domains can be defined using **Propositional STRIPS** (Fikes and Nilsson 1971) where a task is a tuple $P = \langle \mathcal{F}, I, \mathcal{A}, G, \mathcal{C} \rangle$, where

- $\mathcal{F}$ is a set of features (features) and a state s is represented by the features that are true in s
- $I \subseteq \mathcal{F}$ is the initial state
- $G \subseteq \mathcal{F}$ represents the set of goal states, and
- $\mathcal{A}$ is a set of actions.
  - Each action is a triple $a = \langle pre(a), add(a), del(a) \rangle$, which represents the precondition, add, and delete lists respectively, all subsets of $\mathcal{F}$.
  - An action a is applicable in state s if $pre(a) \subseteq s$.
  - If action a is applied in state s, it results in a new state $s' = (s \setminus del(a)) \cup add(a)$.
- $\mathcal{C} : \mathcal{A} \to \mathbb{R}_0^+$ is a cost function that assigns each action a non-negative cost.

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Example: Pick-and-Place in STRIPS

Goal

$$G = \{\, on(green_block,\, blue_block) \,\}$$

Relevant features

- $on(x, y)$: block x is on block y
- $ontable(x)$: block x is on the table
- $clear(x)$: nothing is on x
- $holding(x)$: robot holds x
- $handempty$: nothing being held by the gripper

Example action: **`stack(x,y)`**

$$pre = \{holding(x), clear(y)\}$$

$$add = \{on(x, y), clear(x), handempty\}$$

$$del = \{holding(x), clear(y)\}$$

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Deterministic fully observable environments

The objective is to find a plan $\pi = \langle a_1, \ldots, a_n \rangle$: a sequence of actions that brings an agent from the initial state to a goal state.

The cost $c(\pi)$ of a plan π is $\Sigma_{i=1}^{n}(C(a_i))$. Often, the objective is to find an optimal solution for P: a plan π^* that minimizes the associated cost.

The literature is rich with different approaches developed to solve the planning problem (Bonet and Geffner 2013): more on this later on in the course.

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Representation Complexity of STRIPS

In a planning problem with feature set $\mathcal{F}$, the number of possible states is $2^{|\mathcal{F}|}$. Thus, the state space grows **exponentially** in the number of features.

If transitions are represented explicitly as a graph over states, then in the worst case there can be $2^{|\mathcal{F}|} \times 2^{|\mathcal{F}|} = 2^{2|\mathcal{F}|}$ possible state-transition pairs.

In contrast, the representation complexity of a STRIPS task is the size of its description:

$$O(|\mathcal{F}| + |I| + |G| + \sum_{a \in \mathcal{A}} (|pre(a)| + |add(a)| + |del(a)|))$$

These represent the total number of features and action literals needed to describe the problem.

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Representation Complexity of STRIPS

In the worst case, if no abstraction is captured in the action description (and all transitions are captured explicitly), the complexity of STRIPS and of an explicit transition function is the same.

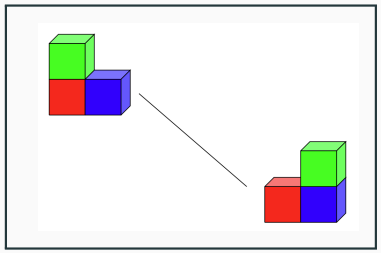
However, in STRIPS, there is a one-to-many mapping from action schema to actions (i.e., possible transitions) so it provides a **compact representation** of planning problems.

Therefore a **small description** may induce a **huge state space**.

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Example: Pick-and-Place in STRIPS - representation complexity

In a transition system encoding of **`stack(x,y)`**, with 3 blocks we have 12 transitions, with 4 blocks we have 84 transitions, and so on...



With STRIPS, we have a single action (for which the representation includes 7 features) regardless of the size of the feature space $|\mathcal{F}|$.

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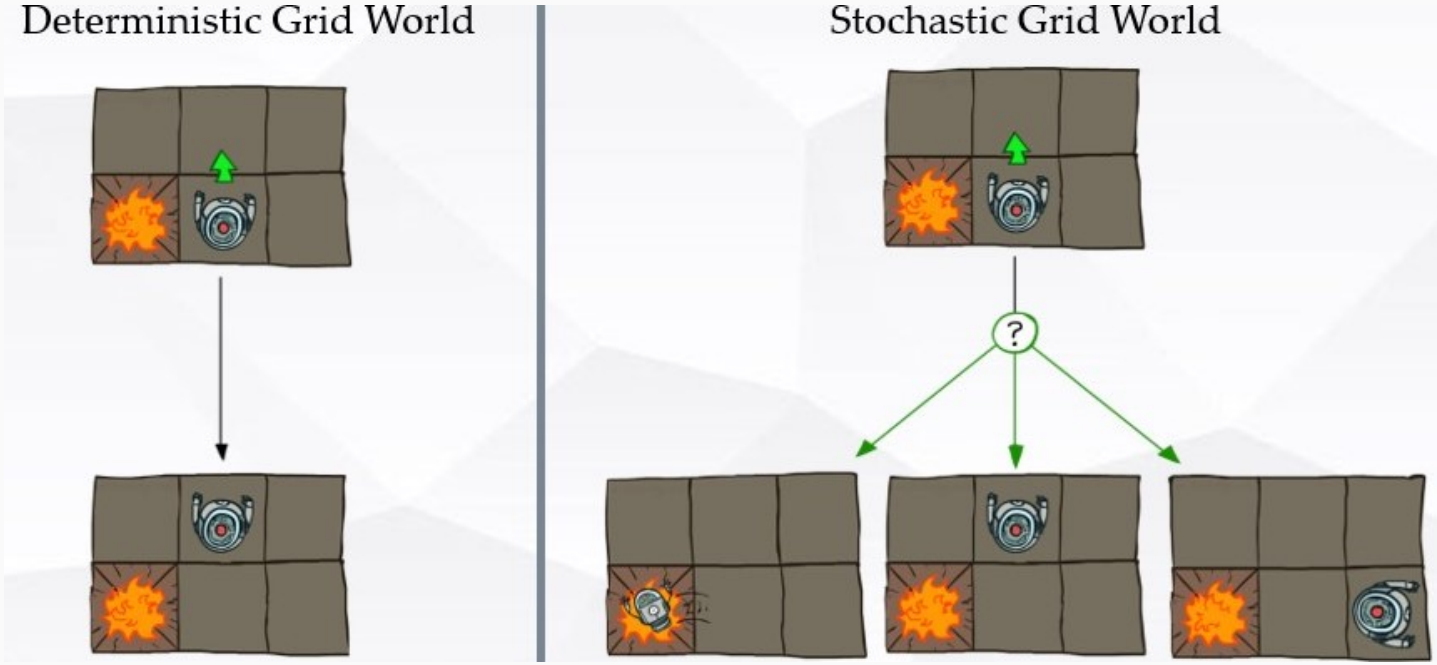
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From Deterministic to Stochastic Domains

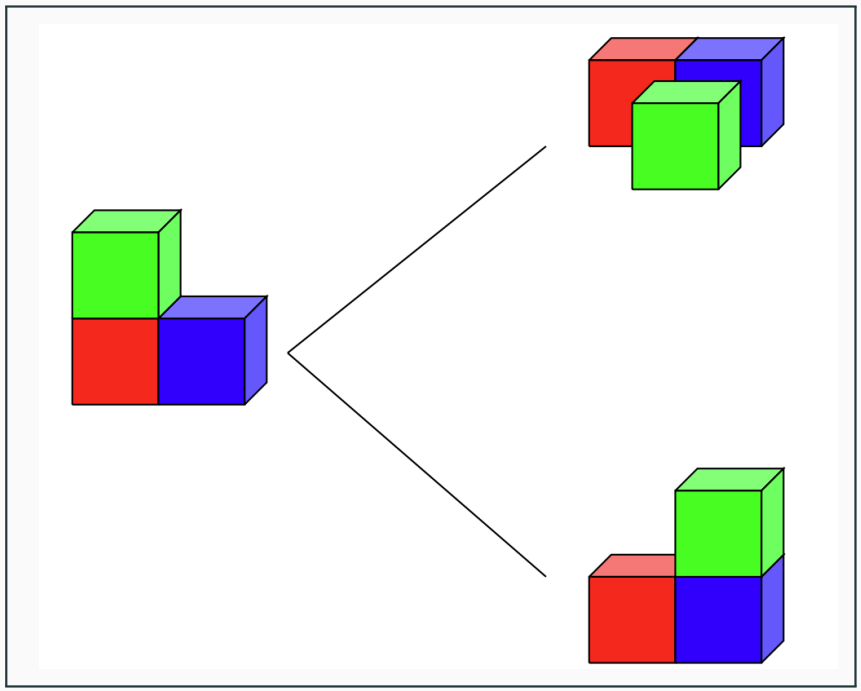


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Nondetermistic dynamics example

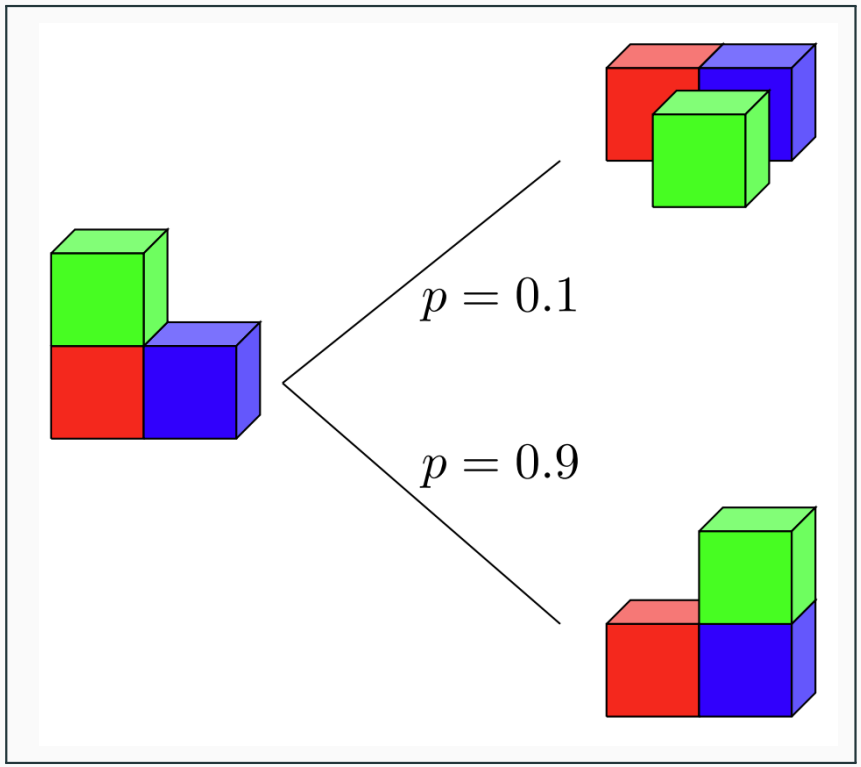
Moving objects with an unreliable robotic hand: move the green block onto the blue block.



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Probabilistic dynamics example

Moving objects with an unreliable robotic hand: move the green block onto the blue block.



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Stochastic fully observable environments

Stochastic fully observable environments are defined using a **Markov Decision Process** (MDP) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where

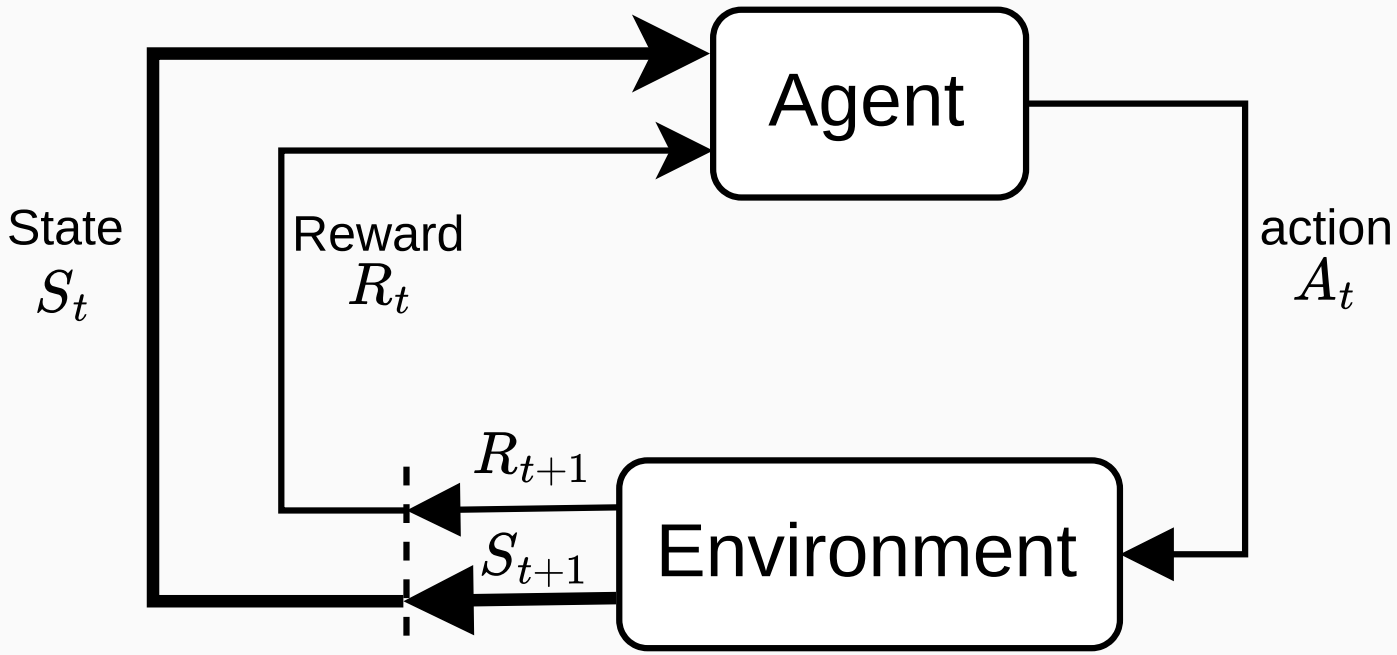
- $\mathcal{S}$ is a finite set of states
- $\mathcal{A}$ is a finite set of actions
- $\mathcal{P}$ is a state transition probability matrix $\mathcal{P}^a_{s,s'} = \mathcal{P}[S_{t+1} = s' | S_t = s, A_t = a]$
- $\mathcal{R}$ is a reward function, $\mathcal{R}^a_s = \mathbb{E}[R_{t+1} | S_t = s, A_t = a]$, and
- **optional:** γ is a discount factor $\gamma \in [0, 1]$ that is used to favor immediate rewards over future rewards.

The Markov property: "The future is independent of the past given the present".

Extensions: Infinite and continuous MDPs, partially observable MDPs, undiscounted, average reward MDPs. etc.

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Stochastic Fully Observable Environments



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Stochastic Fully Observable Environments

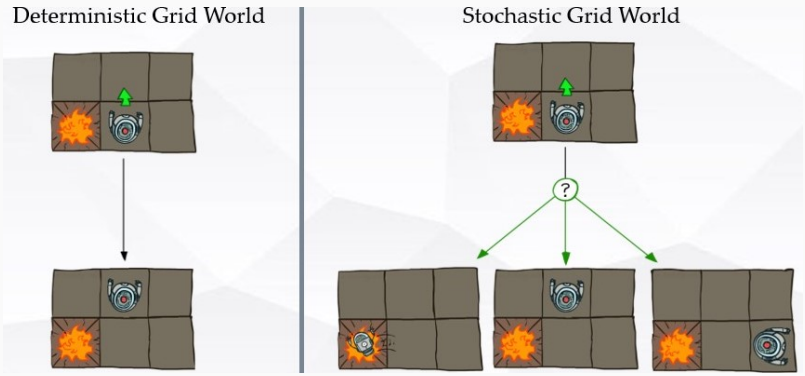
A solution to an MDP is a **policy** that prescribes the action to be taken at every state.

- **Deterministic Policy**: a mapping $\pi : \mathcal{S} \to \mathcal{A}$ from states to actions.
- **Stochastic Policy**: a mapping $\pi : \mathcal{S} \times \mathcal{A} \to [0, 1]$ from state and action pairs to to the probability $\pi(a|s)$ of taking action a when in state s.

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From Deterministic to Stochastic Domains

There is at least one (deterministic) policy that is better than or equal to all other policies. This is an **Optimal Policy** π^*.



If we know there is an optimal deterministic policy for any discounted infinite horizon / finite horizon / stochastic shortest path MDP, why do we need a stochastic policy?

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Factored Representation of Markov Decision Process

Typically, the state space $\mathcal{S}$ is represented using feature space $\mathcal{F}$, such that $s \in d^{|\mathcal{F}|}$, where d is the feature dimension.
The MDP is then $\langle \mathcal{F}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$

The explicit representation of the state space $\mathcal{S}$ is replaced by a factored state representation, where the set of states is described via a set of random variables (features) $\mathcal{F} = f_1, \ldots, f_n$, and where each variable f_i takes on values in some finite domain $Dom(f_i)$.

A state is an assignment of a value $f_i \in Dom(f_i)$ for each variable in $\mathcal{F}$.

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Factored Representation of Markov Decision Process

- Factored MDPs allow representing large, structured MDPs compactly.
- Transition dynamics can then be described compactly, e.g. using a Dynamic Bayesian network (DBN) (Dean and Kanazawa, 1989). This exploits the fact that the transition of a variable often depends only on a small number.
- Momentary rewards can often also be decomposed as a sum of rewards related to individual variables or small clusters of variables

*Boutilier, Craig, Richard Dearden, and Moisés Goldszmidt. "Stochastic dynamic programming with factored representations." Artificial intelligence 121.1-2 (2000): 49-107 and Guestrin, C., Koller, D., Parr, R., and Venkataraman, S. (2003). Efficient solution algorithms for factored MDPs. Journal of Artificial Intelligence Research, 19, 399-468.

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Factored Representations: Key Idea

Instead of modeling transitions between entire states, we model how individual variables evolve.

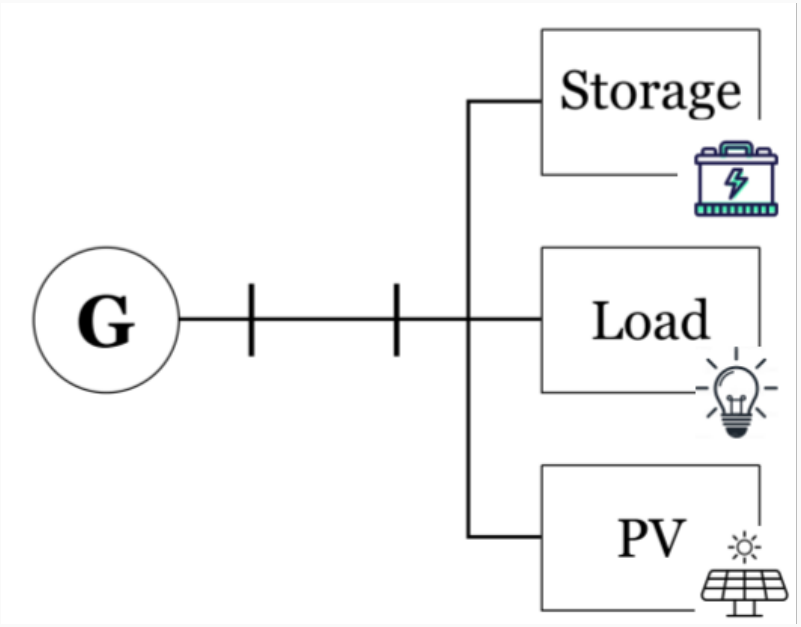
This allows us to represent very large systems

- robots
- energy systems
- logistics networks

without enumerating all states.

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MDP Example: Battery Management



Consider controlling a battery connected to an electricity grid. The energy from the battery can be used to satisfy local demand or be sold to the grid.

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MDP Example: Battery Management (Flat Representation)

States: The state representation includes all relevant variables: state of charge (soc), price, demand: Example state:

$$s_1 = (soc = 60\%,\ price = high,\ demand = medium)$$

Actions: $\mathcal{A} = \{Charge,\ Discharge,\ Idle\}$

Transition model: flat representation must be specified for every triple (s, a, s').

$$\mathcal{P}(s, a, s') = \Pr(S_{t+1} = s' \mid S_t = s, A_t = a)$$

Reward $\mathcal{R}(s, a, s')$

- profit from selling electricity
- cost of purchasing electricity

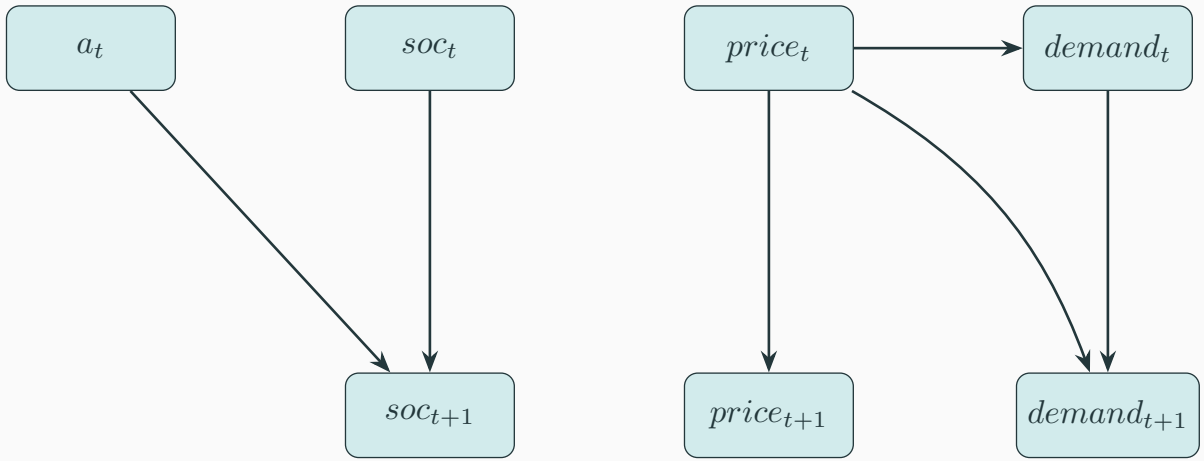
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MDP Example: Battery Management (Factored Representation)

Structured transitions: Instead of specifying $\mathcal{P}(s, a, s')$ for all states, we can encode the dependencies:

- soc_{t+1} depends on (soc_t, a_t)
- $price_{t+1}$ depends on $(price_t)$
- $demand_{t+1}$ depends on $(demand_t, price_t)$

These dependencies can be represented compactly using a Dynamic Bayesian Network.



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Representation Complexity: Flat vs Factored MDP

Total number of states

$$|\mathcal{S}| = 11 \times 3 \times 3 = 99$$

With a **flat MDP representation**, for each state-action pair we must specify a distribution over next states: $O(|\mathcal{S}|^2|\mathcal{A}|)$.

With $|\mathcal{A}| = 3$, we have $99^2 \times 3 = 29{,}403$ transition probabilities.

With a **factored representation**, we only specify transitions for each variable:

- $soc_{t+1} \mid (soc_t, a_t)$
- $price_{t+1} \mid (price_t)$
- $demand_{t+1} \mid (demand_t, price_t)$

Number of parameters is proportional to the number of local dependencies.

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Representation Complexity

In deterministic settings, the state space increased exponentially with the number of features, and the worst-case action space size was induced by the size of the state space

In stochastic settings, transitions must specify a **probability distribution** over successor states. In the worst case, each action may lead to any other state.

If $|\mathcal{S}| = n$ and $|\mathcal{A}| = m$, then an explicit transition model may require specifying $O(mn^2)$ transition probabilities.

In practice, much of modern planing and RL is about coping with **large or continuous state spaces**.

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Properties of the setting: observability

Full observability

Observations/sensing determine the current world state uniquely.

Partial observability

Observations determine current world state only partially: We only know that the current state is one of several possible ones.

No observability

There are no observations to narrow down possible current states. However, can use knowledge of action dynamics to deduce which states we might be in (a.k.a funneling).

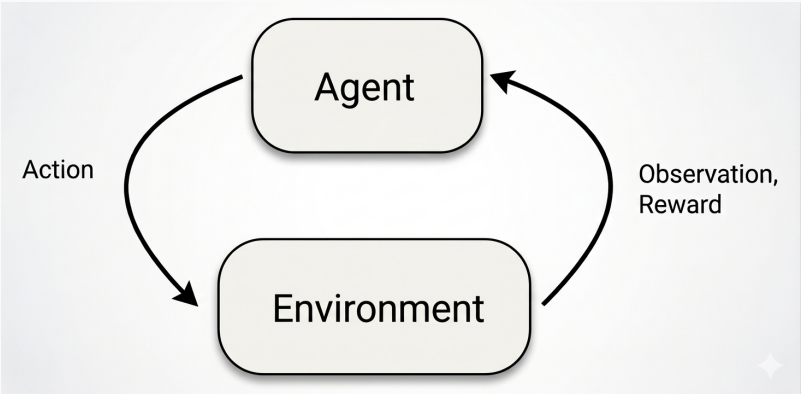
Consequence: If observability is not full, must represent the knowledge an agent has.

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Agency under partial observability

Recall that the environment-agent interactions give rise to a sequence or trajectory: $s_0, a_0, r_1, s_1, a_1, r_2, s_2, a_2, r_3, ...$

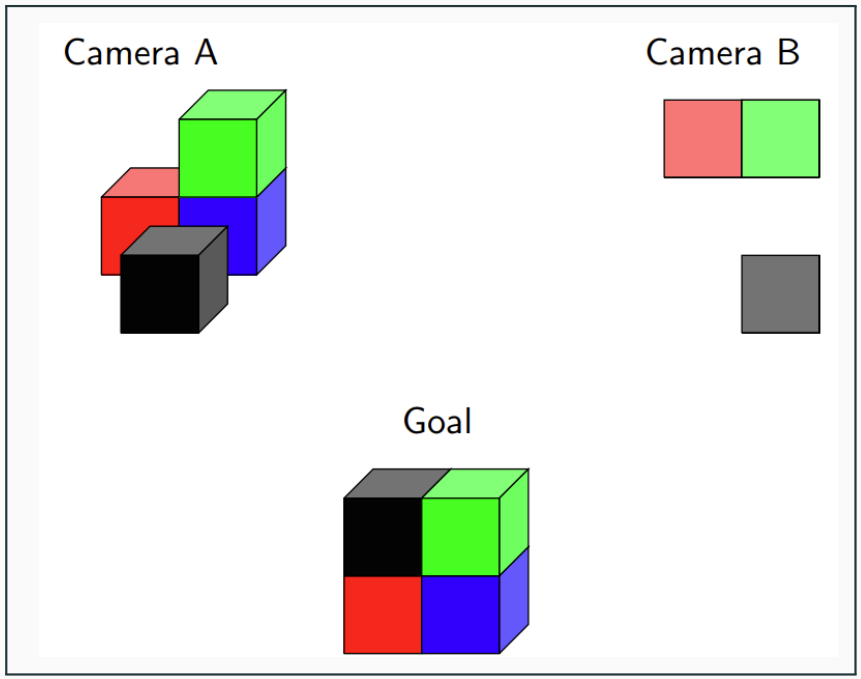
From the point of view of the partially informed agent, we have a **history**: $o_0, a_0, r_1, o_1, a_1, r_2, o_2, a_2, r_3, ...$



The agent needs to maintain its belief based on the current (possibly null) observation.

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Partial Observability - Example



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Partially Observable Markov Decision Process (POMDP)

A **Partially Observable Markov Decision Process**(POMDP) is a tuple $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \Omega, \mathcal{O}, \beta_0 \rangle$ where

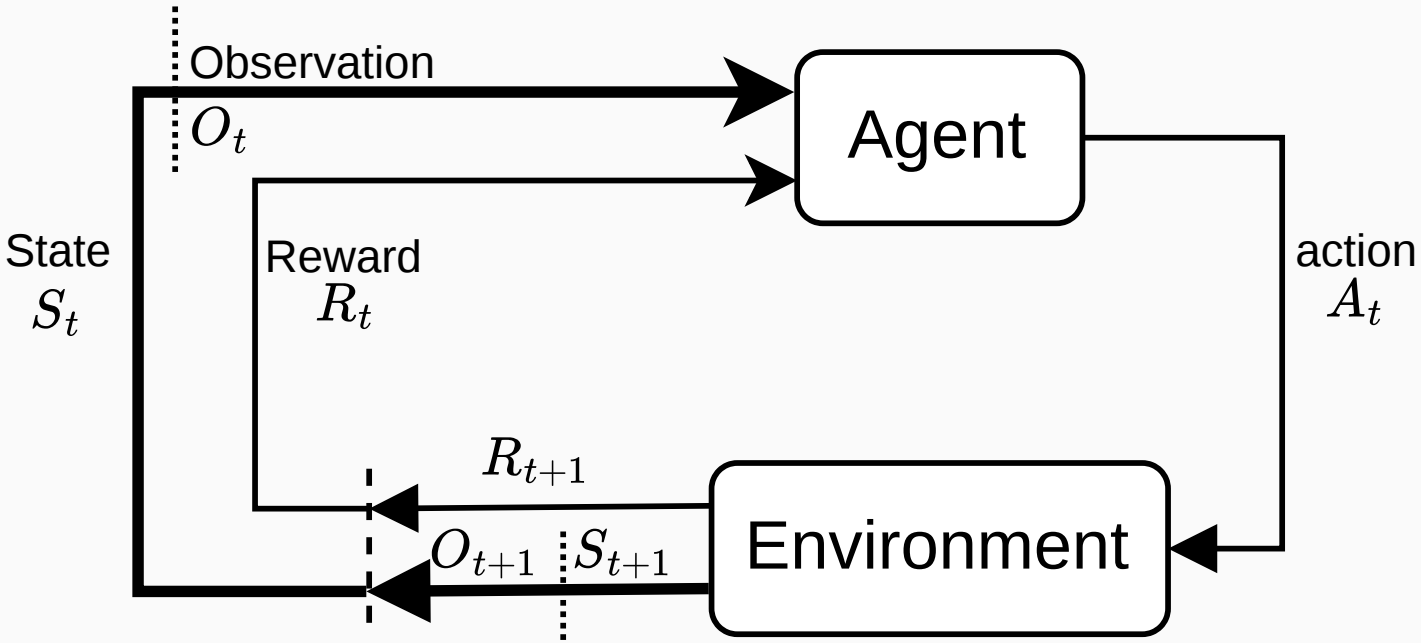
- $\mathcal{S}$, $\mathcal{A}$, $\mathcal{P}$, $\mathcal{R}$ and γ are as for an MDP.
- Ω is a set of observations (observation tokens),
- $\mathcal{O}$ is a sensor function specifying the conditional observation probabilities $\mathcal{O}^o_{s,a} = \mathcal{P}[O_{t+1} = o | S_t = s, A_t = a]$ of receiving observation token $o \in \Omega$ in state s after applying action a [2].
- β_0 the initial belief: a probability distribution over the states such that $\beta_0(s)$ stands for the probability of s being the true initial state.

Markov property still holds

[2] alternatively: $\mathcal{O}^o_s = \mathcal{P}[O_t = o | S_t = s]$

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Representation Complexity

- A POMDP extends an MDP by adding an observation space Ω and an observation model $\mathcal{O}$.
- If $|\mathcal{S}| = n$, $|\mathcal{A}| = m$, and $|\Omega| = k$, then an explicit representation may require:
  - $O(mn^2)$ transition probabilities for $\mathcal{P}$
  - $O(mnk)$ observation probabilities for $\mathcal{O}$
- However, the main increase in complexity comes from the fact that policies are defined over **beliefs**, not states.
- Since beliefs are distributions over $\mathcal{S}$, the belief space is continuous even when $\mathcal{S}$ is finite.

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Summary

- Models for decision-making in AI for single-agent settings
- Limitations and scope of each model

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